

Network Security Agent (Agent 11)

Deep Technical Specification & Operational Manual

Executive Overview & Purpose

The Network Security Agent performs deterministic analysis of network topology data, firewall rule sets, open-port inventories, and exposure lists to identify attack surface, lateral movement paths, and risky perimeter configurations.

1. Architectural Overview & Workflow

- Position in LangGraph Engine: Sole analysis node in Pipeline D (Network Exposure).
- Input Consumption: Accepts nmap-style output, IP:port inventories, firewall rules, and topology descriptions.
- State Propagation: Writes state.network_security_report and appends exposure findings.

2. Exposure & Attack-Surface Analysis

- Open-Port Risk Scoring: Ranks exposed services (e.g., 22/SSH, 3389/RDP, 445/SMB, 1433/MSSQL) by exploitation likelihood.
- Firewall Rule Review: Flags default-deny violations, wildcard source ranges, and management-plane exposure.
- Lateral-Movement Paths: Identifies host pairs where a compromise on one node directly reaches another.

3. Deterministic Segmentation & Hardening Output

- Segmentation Recommendations: Produces allow-list and network-tier suggestions derived from observed flows.
 - Findings Enrichment: Each exposure finding includes severity, affected asset, and concrete hardening step.
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